TBCARE: annotated case report forms for the REDCap class

A fictional study. Use these forms to build the TBCARE project in REDCap step by step; every REDCap feature from the training slides appears at least once.

1. The project

TBCARE follows adults starting tuberculosis (TB) treatment to find out who is most at risk of an unfavourable treatment outcome. Participants are enrolled at two hospitals, **Antwerp (Belgium)** and **Mbarara (Uganda)**, and followed for **12 months**. People with drug-susceptible TB and people with drug-resistant TB receive different regimens and are therefore seen on different schedules, so the REDCap project has **two arms**. The University of Antwerp is the sponsor; funding is from FWO.

	Arm 1: drug-susceptible TB (DS-TB)	Arm 2: drug-resistant TB (DR-TB)
Participants	500 adults	100 adults
Treatment	6-month standard regimen	9-month all-oral regimen
Scheduled visits	Baseline, month 2, month 6, month 12	Baseline, months 1, 2, 4, 6, 9 and 12
Extra monitoring	–	ECG at each treatment visit (heart rhythm, QTcF)
Unscheduled visits	Any number (repeating event)	Any number (repeating event)
Adverse events, medicines	Logged any time (repeating instruments)	Logged any time (repeating instruments)

2. REDCap set-up at a glance

Setting	TBCARE choice	Slides
Project type	Longitudinal, surveys enabled (participants complete consent and quality of life)	21–24
Arms and events	2 arms; scheduled visit events plus an Unscheduled visit event and a Logs event in each arm	26
Repeating	Repeating event: Unscheduled visit. Repeating instruments: Adverse events, Concomitant medications (Logs event)	28
Multi-site	Two Data Access Groups: Antwerp and Mbarara	33
Surveys	e-Consent (baseline); Quality of life in English and Runyankole (Multi-Language Management)	32
Quality	Validation, branching, calculated fields, Data Quality rules, queries, record locking	16–18, 34–36
Exports	De-identified: identifier fields removed, dates shifted, record ID hashed	42

3. Arms and events

Days offset and windows are entered in **Define My Events**; REDCap uses them for the schedule and the Calendar. The Unscheduled and Logs events have no fixed date.

Arm 1 (DS-TB) event	Days offset	Window	Unique event name	Arm 2 (DR-TB) event	Days offset	Window	Unique event name
Baseline	0	0	baseline_arm_1	Baseline	0	0	baseline_arm_2
Month 2	60	± 7	month_2_arm_1	Month 1	30	± 7	month_1_arm_2
Month 6 (end of treatment)	180	± 14	month_6_arm_1	Month 2	60	± 7	month_2_arm_2
Month 12	365	± 14	month_12_arm_1	Month 4	120	± 14	month_4_arm_2
Unscheduled visit (repeating)	–	–	unscheduled_arm_1	Month 6	180	± 14	month_6_arm_2
Logs	–	–	logs_arm_1	Month 9 (end of treatment)	270	± 14	month_9_arm_2
				Month 12	365	± 14	month_12_arm_2
				Unscheduled visit (repeating)	–	–	unscheduled_arm_2
				Logs	–	–	logs_arm_2

4. Which form is collected at which event (Designate Instruments)

✓ = collected at that event · R = repeating instrument (as many as needed) · S = completed by the participant as a survey

Form	B	M2	M6	M12	Unsch.	Logs	|	B	M1	M2	M4	M6	M9	M12	Unsch.	Logs
F1 Screening and eligibility	✓							✓								
F2 e-Consent	S							S								
F3 Enrolment and demographics	✓							✓								
F4 Clinical assessment	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	
F5 Laboratory results	✓	✓	✓		✓			✓	✓	✓	✓	✓	✓		✓	
F6 ECG monitoring (arm 2 only)								✓	✓	✓	✓	✓	✓			
F7 Quality of life	S		S	S				S				S		S		
F8 Adverse events						R										R
F9 Concomitant medications						R										R
F10 Treatment outcome			✓										✓			
F11 Study completion				✓										✓		

Left block: **Arm 1 (DS-TB)**. Right block: **Arm 2 (DR-TB)**. B = baseline; M = month; Unsch. = Unscheduled visit (a repeating event containing F4 and F5).

5. The forms

Legend: **Identifier** = tick “Identifier? yes” (removed in de-identified exports) · **Required** = required field · action tags (e.g., @TODAY) go in the Field Annotation · codes are written “1, Yes | 0, No”, as in the data dictionary.

F1 Screening and eligibility

Completed by: site staff · Event: Baseline (both arms) · Slides: 15–18

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Study ID	record_id	text	–	Auto-numbered (e.g., 1-001 at Antwerp, 2-001 at Mbarara). Never contains names
2	Date of screening	screen_date	text	–	date_dmy; @TODAY; **Required**
3	Is the person 18 years or older?	age_18	yesno	1, Yes | 0, No	**Required**
4	Is TB confirmed?	tb_confirmed	radio	1, Bacteriologically confirmed | 2, Clinically diagnosed | 0, Not confirmed	**Required**
5	Type of TB	tb_type	radio	1, Drug-susceptible | 2, Drug-resistant	Decides the arm: create the record in arm 1 or arm 2
6	Has written informed consent been given?	consent_ok	yesno	1, Yes | 0, No	**Required**
7	Eligible	eligible	calc	–	if([age_18]='1' and [tb_confirmed]<>'0' and [consent_ok]='1', 1, 0)
8	Reason not eligible	not_eligible_why	notes	–	Branching: [eligible] = '0'

F2 e-Consent (survey)

Completed by: participant, on a tablet · Event: Baseline · Survey with the e-Consent framework · Slide: 32

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Information sheet	info_text	descriptive	–	Shows the information sheet (version 2.0, 01-09-2026)
2	I agree to take part in TBCARE	cons_part	yesno	1, Yes | 0, No	**Required**
3	I agree that my data are stored for 10 years	cons_store	yesno	1, Yes | 0, No	**Required**
4	I agree that de-identified data may be shared with other researchers	cons_share	yesno	1, Yes | 0, No	Drives what may be shared (DMP section 5)
5	I agree that my TB samples may be sequenced	cons_seq	yesno	1, Yes | 0, No	Links to the Nagoya question in Part 2
6	Full name	cons_name	text	–	**Identifier**
7	Signature	cons_sign	file (signature)	–	**Identifier**; e-Consent stores a signed PDF with version and date

F3 Enrolment and demographics

Completed by: site staff · Event: Baseline · Slides: 15–18

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Date of enrolment	enrol_date	text	–	date_dmy; @TODAY; **Required**
2	First name	first_name	text	–	**Identifier**
3	Last name	last_name	text	–	**Identifier**
4	Date of birth	dob	text	–	date_dmy; **Identifier**; min 01-01-1920, max today
5	Age at enrolment (years)	age	calc	–	rounddown(datediff([dob], [enrol_date], 'y'), 0)
6	Sex at birth	sex	radio	1, Male | 2, Female	**Required**
7	Currently pregnant?	pregnant	yesno	1, Yes | 0, No	Branching: [sex] = '2' and [age] >= 15 and [age] <= 49
8	Mobile phone number	phone	text	–	phone validation; **Identifier**
9	District of residence	district	dropdown	List of districts per country	Indirect identifier: keep district, not village
10	Highest education	education	radio	0, None | 1, Primary | 2, Secondary | 3, Higher | 99, Prefer not to say	
11	HIV status	hiv_status	radio	1, Positive | 0, Negative | 9, Unknown	**Required**
12	On antiretroviral therapy?	on_art	yesno	1, Yes | 0, No	Branching: [hiv_status] = '1'
13	Previous TB treatment?	prev_tb	yesno	1, Yes | 0, No	

F4 Clinical assessment

Completed by: site staff · Events: every scheduled visit and every Unscheduled visit · Slides: 16–18, 31

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Visit date	visit_date	text	–	date_dmy; @TODAY; **Required**; Data Quality rule: not before enrolment
2	Vital signs	vitals_table	descriptive	–	Field embedding: a table showing {weight_kg} {height_cm} {temp_c}
3	Weight (kg)	weight_kg	text	–	number_1dp; min 25, max 200; **Required**
4	Height (cm)	height_cm	text	–	number; min 120, max 220; Branching: [event-name] = 'baseline_arm_1' or [event-name] = 'baseline_arm_2'
5	BMI (kg/m²)	bmi	calc	–	round([weight_kg] / ([baseline_arm_1][height_cm]/100)^2, 1) (use baseline_arm_2 in arm 2)
6	Temperature (°C)	temp_c	text	–	number_1dp; min 34, max 42
7	Symptoms in the last 2 weeks	symptoms	checkbox	1, Cough | 2, Fever | 3, Night sweats | 4, Weight loss | 5, Coughing blood | 6, None	@NONEOFTHEABOVE='6'
8	Duration of cough (weeks)	cough_weeks	text	–	integer; min 0, max 104; Branching: [symptoms(1)] = '1'
9	Doses missed in the last 30 days	doses_missed	text	–	integer; min 0, max 30; Branching: hidden in the Baseline events
10	Adherence (%)	adherence_pct	calc	–	round(100 * (30 - [doses_missed]) / 30, 0)
11	Reason for unscheduled visit	unsched_reason	notes	–	Branching: only in the Unscheduled visit event ([event-name] = 'unscheduled_arm_1' or 'unscheduled_arm_2')

F5 Laboratory results

Completed by: site staff (or monthly import from the hospital laboratory) · Slides: 15, 17

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Was a TB test done at this visit?	lab_eval	yesno	1, Yes | 0, No	**Required**
2	Date of specimen	spec_date	text	–	date_dmy; Branching: [lab_eval] = '1'
3	Tests done	test_type	checkbox	1, Smear microscopy | 2, GeneXpert | 3, Culture	Branching: [lab_eval] = '1'
4	Smear result	smear	dropdown	0, Negative | 1, Scanty | 2, 1+ | 3, 2+ | 4, 3+	Branching: [test_type(1)] = '1'
5	GeneXpert result	xpert	radio	0, MTB not detected | 1, MTB detected, RIF resistance not detected | 2, MTB detected, RIF resistance detected | 9, Invalid	Branching: [test_type(2)] = '1'
6	Culture result	culture	radio	0, Negative | 1, Positive | 2, Contaminated	Branching: [test_type(3)] = '1'
7	Isolate shipped to Antwerp for sequencing?	isolate_ship	yesno	1, Yes | 0, No	Only if [culture] = '1' and consent [cons_seq] = '1' (cross-event branching)

F6 ECG monitoring (arm 2 only)

Completed by: site clinician · Events: arm 2 treatment visits · Shows an arm-specific form · Slides: 27, 32

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Date of ECG	ecg_date	text	–	date_dmy; **Required**
2	QTcF interval (ms)	qtcf	text	–	integer; min 300, max 700
3	QTcF above 500 ms	qtcf_high	calc	–	if([qtcf] > 500, 1, 0); Alert: e-mail the site clinician and data manager when 1
4	Action taken	ecg_action	radio	0, None | 1, Repeat ECG | 2, Drug stopped	Branching: [qtcf_high] = '1'

F7 Quality of life (survey)

Completed by: participant · Events: Baseline, month 6, month 12 · English and Runyankole (Multi-Language Management) · Slides: 15, 32

A short, study-specific questionnaire written for this exercise (not a validated instrument).

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Over the last week, how much has your health limited your daily activities?	qol_1	radio	1, Not at all | 2, A little | 3, Moderately | 4, A lot | 5, Completely	Same scale for qol_1 to qol_4 (use a matrix group)
2	…how much pain or discomfort have you had?	qol_2	radio	same scale	Matrix group: qol
3	…how worried or low have you felt?	qol_3	radio	same scale	Matrix group: qol
4	…how much has TB affected your work or income?	qol_4	radio	same scale	Matrix group: qol
5	How good is your health today?	qol_vas	slider	0 = worst, 100 = best	Slider with number shown
6	Anything else you would like to tell us?	qol_comment	notes	–	Free text; check before sharing (may contain identifiers)

F8 Adverse events (repeating instrument)

Completed by: site staff · Event: Logs · One instance per adverse event · Slides: 28, 32

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Adverse event (description)	ae_term	text	–	**Required**
2	Date of onset	ae_onset	text	–	date_dmy; **Required**
3	Severity grade	ae_grade	radio	1, Mild | 2, Moderate | 3, Severe | 4, Life-threatening | 5, Death	
4	Serious adverse event?	ae_serious	yesno	1, Yes | 0, No	Alert: e-mail the PI within 24 h when [ae_serious] = '1'
5	Related to TB treatment?	ae_related	radio	1, Related | 2, Possibly related | 0, Not related	
6	Outcome	ae_outcome	radio	1, Resolved | 2, Ongoing | 3, Resolved with sequelae | 4, Fatal	

F9 Concomitant medications (repeating instrument)

Completed by: site staff · Event: Logs · One instance per medicine · Slide: 28

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Medicine name	med_name	text	–	**Required**
2	Reason for use	med_reason	text	–	
3	Start date	med_start	text	–	date_dmy
4	Still taking it?	med_ongoing	yesno	1, Yes | 0, No	
5	Stop date	med_stop	text	–	date_dmy; Branching: [med_ongoing] = '0'; Data Quality rule: not before start date

F10 Treatment outcome

Completed by: site clinician · Event: end of treatment (month 6 in arm 1, month 9 in arm 2) · Slides: 17, 36

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Date of outcome	outcome_date	text	–	date_dmy; **Required**
2	Treatment outcome	tx_outcome	radio	1, Cured | 2, Treatment completed | 3, Treatment failed | 4, Died | 5, Lost to follow-up | 6, Not evaluated	WHO outcome categories; **Required**
3	Date of death	death_date	text	–	date_dmy; Branching: [tx_outcome] = '4'
4	Treatment success	tx_success	calc	–	if([tx_outcome] = '1' or [tx_outcome] = '2', 1, 0)
5	Form verified against source	verified	yesno	1, Yes | 0, No	Then lock the form (record locking + e-signature)

F11 Study completion

Completed by: site staff · Event: month 12 · Slide: 36

#	Question (field label)	Variable	Type	Choices (code, label)	Validation, logic, notes
1	Did the participant complete 12 months of follow-up?	completed	yesno	1, Yes | 0, No	**Required**
2	Reason for not completing	withdraw_reason	radio	1, Withdrew consent | 2, Lost to follow-up | 3, Died | 4, Moved away | 5, Other	Branching: [completed] = '0'
3	Date of last contact	last_contact	text	–	date_dmy

6. Rules, roles and exports

Data Quality rules (custom)

Rule	Logic	Slide
Pregnant male	[sex] = '1' and [pregnant] = '1'	35
Visit before enrolment (arm 1)	datediff([baseline_arm_1][enrol_date], [visit_date], 'd', true) < 0	35
Medicine stopped before it started	datediff([med_start], [med_stop], 'd', true) < 0	35
Death without a date	[tx_outcome] = '4' and [death_date] = ''	35
Weight change above 10 kg since baseline (arm 1)	abs([weight_kg] - [baseline_arm_1][weight_kg]) > 10	35

Alerts and notifications

Trigger	Who is informed	Slide
[ae_serious] = '1'	PI and data manager, by e-mail, immediately	32
[qtcf_high] = '1'	Site clinician and data manager	32
Quality of life survey not completed 7 days after the visit	Participant (reminder), via the participant list	32

User roles and Data Access Groups

Role	DAG	Rights	Slides
Site data entry	Antwerp or Mbarara	Enter and edit own site; respond to queries; no export	33, 40
Site clinician	Antwerp or Mbarara	As data entry, plus lock and e-sign F10	36, 40
Data manager	none (sees all)	All sites; Data Quality; open and close queries; full export	34, 40
Statistician	none	De-identified export only	40, 42
PI	none	View all; user rights; full export	40

De-identified export for the statistician

Identifier fields removed: first_name, last_name, dob, phone, cons_name, cons_sign. Free-text fields removed: not_eligible_why, unsched_reason, qol_comment. Dates shifted by the same random number of days per record; record ID hashed. Age (calculated) and district are kept. Slide 42.

7. Where each REDCap feature appears in TBCARE

REDCap feature	In TBCARE	Slides
Project lifecycle (Development to Production)	Build and test with fake records 1-001 to 1-003 before going live	8, 44
Data dictionary	All 11 forms; variable names as in the tables above	14
Field types	text, notes, radio, dropdown, checkbox, yes/no, calc, slider, file (signature), descriptive	15
Validation, required, identifiers	Dates, weight, temperature, QTcF; identifiers in F2 and F3	16
Branching logic (simple, checkbox, cross-event)	pregnant; smear/Xpert/culture; isolate_ship	17
Calculated fields, action tags, field embedding	age, bmi, adherence_pct, eligible, tx_success; @TODAY, @HIDDEN, @NONEOFTHEABOVE; vitals table	18
Multi-arm longitudinal project	Arm 1 DS-TB, arm 2 DR-TB, different schedules; F6 only in arm 2	21–27
Repeating event and repeating instruments	Unscheduled visit; Adverse events, Concomitant medications	28
Scheduling and Calendar	Days offset and windows per event	29
Surveys, e-Consent, MLM, alerts	F2, F7; alerts on SAE and QTcF	32
Data Access Groups	Antwerp, Mbarara	33
Queries, Data Quality rules, locking	Section 6 rules; lock F10 after verification	34–36
User rights, logging, exports, API	Section 6 roles; de-identified export; REDCapR for the statistician	40–43